

A new Potential-Based Reward Shaping for Reinforcement Learning Agent

Babak Badnava
Nasser Mozayani

School of Computer Engineering
Iran University of Science and Technology

November 2017

Abstract

Potential-based reward shaping (PBRS) is a particular category of machine learning methods which aims to improve the learning speed of a reinforcement learning agent by extracting and utilizing extra knowledge while performing a task. There are two steps in the process of transfer learning: extracting knowledge from previously learned tasks and transferring that knowledge to use it in a target task. The latter step is well discussed in the literature with various methods being proposed for it, while the former has been explored less. With this in mind, the type of knowledge that is transmitted is very important and can lead to considerable improvement. Among the literature of both the transfer learning and the potential-based reward shaping, a subject that has never been addressed is the knowledge gathered during the learning process itself. In this paper, we presented a novel potential-based reward shaping method that attempted to extract knowledge from the learning process. The proposed method extracts knowledge from episodes' cumulative rewards. The proposed method has been evaluated in the Arcade learning environment and the results indicate an improvement in the learning process in both the single-task and the multi-task reinforcement learner agents.

Keywords : potential-based reward shaping, reinforcement learning, reward shaping, knowledge extraction

1 Introduction

In reinforcement learning (RL) problems, an agent learns to maximize a reward signal, which may be time-delayed [16]. The RL framework has gained much success in handling complex problems in last years. However, mastering difficult tasks is often slow. Thus most of the RL researcher focuses on improving the speed of the learning process by using the knowledge that provided by an expert or a heuristic function.

Transfer learning (TL) is one of the approaches that try to improve the speed of learning by using knowledge extracted from previously learned tasks [6]. Reward shaping (RS) is also one of the methods that have been used for transferring such knowledge or any other type of knowledge.

In problems that could be modeled using RL, many tasks have kind of sparse reward signal. For example, there are some tasks that an agent will receive nothing until it gets to a goal or it gets nothing until some events occur in the environment. The reward function of a task in an

一种新的基于潜力的强化学习代理奖励塑造

巴巴克·巴德纳
瓦·纳赛尔·莫扎
亚尼

伊朗科技大学计算机工程学院

2017年11月

Abstract

基于电位的奖励塑造 (PBRs) 是一类特殊的机器学习方法, 旨在通过在执行任务时提取和利用额外的知识来提高强化学习代理的学习速度。迁移学习的过程有两个步骤: 从先前学习的任务中提取知识, 并将该知识迁移到目标任务中。后一步在文献中得到了很好的讨论, 并提出了各种方法, 而前一步的探索较少。考虑到这一点, 所传播的知识类型非常重要, 并且可以带来相当大的进步。在迁移学习和基于潜力的奖励塑造的文献中, 从未讨论过的一个主题是在学习过程本身中收集的知识。在本文中, 我们提出了一种新颖的基于潜力的奖励塑造方法, 试图从学习过程中提取知识。所提出的方法从情节的累积奖励中提取知识。所提出的方法已在 Arcade 学习环境中进行了评估, 结果表明单任务和多任务强化学习代理的学习过程都有所改进。

Keywords : 基于潜力的奖励塑造、强化学习、奖励塑造、知识提取

1 Introduction

在强化学习 (RL) 问题中, 代理学习最大化奖励信号, 这可能是有时间延迟的[16]。近年来, 强化学习框架在处理复杂问题方面取得了巨大成功。然而, 掌握困难的任务往往很慢。因此, 大多数强化学习研究人员专注于通过使用专家或启发式函数提供的知识来提高学习过程的速度。

迁移学习 (TL) 是尝试通过使用从先前学习的任务中提取的知识来提高学习速度的方法之一[6]。奖励塑造 (RS) 也是用于转移此类知识或任何其他类型知识的方法之一。

在可以使用强化学习建模的问题中, 许多任务都有稀疏的奖励信号。例如, 在某些任务中, 代理在达到目标之前不会收到任何信息, 或者在环境中发生某些事件之前不会收到任何信息。任务的奖励函数

environment is independent of time, and the agent tries to maximize the sum of its discounted reward in an episode, while from an episode to another episode of learning, the agent could use a knowledge extracted from past episodes to reinforce the reward signal. There is much information that can be used to reinforce the reward signal. For example, after one episode of learning has done, the agent could compare its performance with the best episode of the learning or the worst one, and by this comparison, it could improve the learning process of itself.

2 Background

2.1 Reinforcement learning

RL is a group of machine learning methods, which can be used for training an agent using environmental feedback. An RL agent in an environment can be modeled as a Markov decision process (MDP) [16]. An MDP is a tuple like (S, A, T, γ, R) , and in this tuple: S is a set of states that appear in the environment; A is a set of agent's possible actions; $T : S \times A \times S \rightarrow [0, 1]$ is a function which gives probability of reaching state s' while agent currently is in state s and chose action a to perform; $R : S \times A \times S \rightarrow \mathbb{R}$ is a function which gives amount of reward the agent will receives when performed the action a in state s and transferred to state s' ; γ called discount factor which indicates how future rewards are important for the agent.

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \quad (1)$$

The agent wants to maximize equation 1, which is called the expected discounted return, during its lifetime [16]. There are many methods for training RL agents. Methods such as Q-learning[19] and SARSA which estimate a value for each $(state, action)$ pair and by using this estimated value compute a policy that can maximize the agent's discounted return. There are also some methods that use policy gradient [2, 17, 21] to approximate a policy and use this policy for the agent's decision making.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right] \quad (2)$$

2.2 Reward shaping

Reward shaping (RS) refers to allowing an agent to train on an artificial reward signal rather than environmental feedback [18]. However, this artificial signal must be a potential function. Otherwise, the optimal policy will be different for the new MDP [14]. Authors of [14] prove that F is a potential function if there exists a real-valued function $\phi : S \rightarrow \mathbb{R}$ such that for all $s \in S - \{s_0\}, a \in A, s' \in S$:

$$F(s, s') = \gamma \phi(s') - \phi(s) \quad (3)$$

Consequently by using this potential function, the optimal policy in the new MDP ($\mathcal{M}' = (S, A, T, \gamma, R + F)$) remains optimal for the old MDP ($\mathcal{M} = (S, A, T, \gamma, R)$). In the new MDP, equation 3, and all of the extended equations from equation 3, γ is unchanged and must have the same value as in the old MDP. There are also other works in the literature that extend the potential function. Authors of [20] have shown that the potential function can be a function of the joint state-action space. Next, Authors of [7] have shown that the potential function could be a dynamic function and that might change over the time while all the properties of potential based reward shaping (PBRs) remain

环境与时间无关，代理试图在一个情节中最大化其折扣奖励的总和，而从一个情节到另一个情节的学习，代理可以使用从过去的情节中提取的知识来强化奖励信号。有很多信息可以用来强化奖励信号。例如，在完成一个阶段的学习后，智能体可以将其表现与学习中最好的阶段或最差的阶段进行比较，通过这种比较，它可以改进自身的学习过程。

2 Background

2.1 Reinforcement learning

强化学习是一组机器学习方法，可用于使用环境反馈来训练代理。环境中的强化学习代理可以建模为马尔可夫决策过程（MDP）[16]。一个MDP是一个类似 (S, A, T, γ, R) 的元组，在这个元组中： S 是环境中出现的一组状态； A 是一组代理可能的动作； $T: S \times A \times S \rightarrow [0, 1]$ 是一个函数，给出当智能体当前处于状态 s 并选择要执行的动作 a 时达到状态 s' 的概率； $R: S \times A \times S \rightarrow \mathbb{R}$ 是一个函数，它给出了代理在状态 s 下执行动作 a 并转移到状态 s' 时将收到的奖励金额； γ 称为折扣因子，表明未来奖励对于代理有多重要。

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \quad (1)$$

代理希望在其生命周期内最大化方程 1，即预期贴现收益 [16]。训练 RL 智能体的方法有很多种。Q-learning[19] 和 SARSA 等方法估计每个 $(state, action)$ 对的值，并使用该估计值计算可以最大化代理折扣回报的策略。还有一些方法使用策略梯度[2,17,21]来近似策略并使用该策略进行代理决策。

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right] \quad (2)$$

2.2 Reward shaping

奖励塑造（RS）是指允许代理根据人工奖励信号而不是环境反馈进行训练[18]。然而，这个人工信号一定是一个潜在的函数。否则，新 MDP 的最优策略将有所不同 [14]。[14] 的作者证明，如果存在实值函数 $\phi: S \rightarrow \mathbb{R}$ 使得对于所有 $s \in S - \{s_0\}, a \in A, s' \in S$ ，则 F 是一个势函数：

$$F(s, s') = \gamma \phi(s') - \phi(s) \quad (3)$$

因此，通过使用这个势函数，新 MDP($\mathcal{M}' = (S, A, T, \gamma, R + F)$) 中的最优策略对于旧 MDP($\mathcal{M} = (S, A, T, \gamma, R)$) 仍然是最优的。在新 MDP、方程 3 以及方程 3 的所有扩展方程中， γ 保持不变，并且必须具有与旧 MDP 中相同的值。文献中还有其他作品扩展了潜在功能。[20] 的作者表明，势函数可以是联合状态-动作空间的函数。接下来，[7] 的作者表明，潜在函数可能是一个动态函数，并且可能会随着时间的推移而变化，而基于潜在的奖励塑造（PBRs）的所有属性仍然保留

constant. In addition to [20] and [7], [12] combines these two extensions and show that any function in the form of equation 6 could be a potential function.

$$F(s, a, s', a') = \gamma \phi(s', a') - \phi(s, a) \quad (4)$$

$$F(s, t, s', t') = \gamma \phi(s', t') - \phi(s, t) \quad (5)$$

$$F(s, a, t, s', a', t') = \gamma \phi(s', a', t') - \phi(s, a, t) \quad (6)$$

By using reward shaping Q-learning update rule will turn into equation 7. In equation 7, F could be any of equations represented in 3, 4, 5, or 6. Equation 7 will give us this opportunity to enhance the learning by providing some extra information about the problem for the agent. This additional knowledge can come from any source like human knowledge of the problem, some heuristic function, or the knowledge that extracted by the agent.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_{t+1} + F + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right] \quad (7)$$

3 Related works

The first method, which should be considered, is the Multi-grid RL with RS, [11] which propose a method for learning a potential function in an online manner. Authors of [11] estimate a value function during the learning process and by using this value function shape a potential function. The estimated value function is the value of an abstract state. State abstraction could be applied by any method.

$$V(z) = (1 - \alpha_v) V(z) + \alpha_v \left(r_v + \gamma_v^t(z') \right) \quad (8)$$

$$F(s, s') = \gamma_v V(z') - V(z) \quad (9)$$

Another work that has been done in the literature is [10], which proposed a potential function based on a plan. In [10], the agent gets an extra reward based on progress in the plan. In equation 10, z is an abstract state that can be any state of the plan, and the function $step(z)$ returns the time step at which given the abstract state z appears during the process of executing the plan.

$$\phi(s) = step(z) \quad (10)$$

Another work that is an extended version of [10], for multi-agent reinforcement learning, is [8]. In [8] two methods have been proposed, which can be used to shape the reward signal of agents and improve their learning process.

Another work that has been presented in the literature of transfer learning is a method that transfers the agent's policy using reward shaping. Authors of [5] assume that there is a mapping function, which maps target task to source task. By using this mapping function, they define a potential function for shaping the reward signal. The proposed potential function as shown in equation 11 is defined based on the policy of source task. In equation 11, $\mathcal{X}_S$ is a mapping function from target task state space to source task state space, and $\mathcal{X}_A$ is a mapping function from target task action space to source task action space, and π is the policy of agent in the source task.

$$\phi(s, a) = \pi(\mathcal{X}_S(s), \mathcal{X}_A(a)) \quad (11)$$

持续的。除了[20]和[7]之外，[12]结合了这两个扩展并表明方程6形式的任何函数都可以是势函数。

$$F(s, a, s', a') = \gamma \phi(s', a') - \phi(s, a) \quad (4)$$

$$F(s, t, s', t') = \gamma \phi(s', t') - \phi(s, t) \quad (5)$$

$$F(s, a, t, s', a', t') = \gamma \phi(s', a', t') - \phi(s, a, t) \quad (6)$$

通过使用奖励塑造，Q-学习更新规则将变成等式 7。在等式 7 中， F 可以是 3、4、5 或 6 中表示的任何等式。等式 7 将为我们提供通过为代理提供一些有关问题的额外信息来增强学习的机会。这些额外的知识可以来自任何来源，例如人类对问题的了解、某些启发式函数或代理提取的知识。

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[R_{t+1} + F + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right] \quad (7)$$

3 Related works

第一种应该考虑的方法是带有 RS 的多网格强化学习，[11]，它提出了一种以在线方式学习势函数的方法。[11] 的作者在学习过程中估计了一个价值函数，并通过使用这个价值函数塑造了一个潜在的函数。估计值函数是抽象状态的值。状态抽象可以通过任何方法应用。

$$V(z) = (1 - \alpha_v) V(z) + \alpha_v (r_v + \gamma_v^t(z')) \quad (8)$$

$$F(s, s') = \gamma_v V(z') - V(z) \quad (9)$$

文献中已经完成的另一项工作是[10]，它提出了基于计划的势函数。在[10]中，智能体根据计划的进展获得额外的奖励。在方程10中， z 是一个抽象状态，可以是计划的任何状态，函数 $step(z)$ 返回在执行计划的过程中给定抽象状态 z 出现的时间步长。

$$\phi(s) = step(z) \quad (10)$$

另一项针对多智能体强化学习的工作是[10]的扩展版本，[8]。在[8]中提出了两种方法，可用于塑造智能体的奖励信号并改善其学习过程。

迁移学习文献中提出的另一项工作是一种使用奖励塑造来迁移代理策略的方法。[5]的作者假设存在一个映射函数，它将目标任务映射到源任务。通过使用这个映射函数，他们定义了一个用于塑造奖励信号的潜在函数。方程 11 所示的拟议势函数是根据源任务的策略定义的。在等式11中， $\mathcal{X}_S$ 是从目标任务状态空间到源任务状态空间的映射函数， $\mathcal{X}_A$ 是从目标任务动作空间到源任务动作空间的映射函数， π 是源任务中代理的策略。

$$\phi(s, a) = \pi(\mathcal{X}_S(s), \mathcal{X}_A(a)) \quad (11)$$

In [5], RS has been used as a knowledge-transfer procedure, and a learned policy from another task has been used as knowledge.

There are many other works in the literature that proposed a potential based reward shaping method for multi-agent reinforcement learning (MARL) and single-agent reinforcement learning (SARL). [4] proposed a method for RS, while there is a demonstration of a task. [4] proposed a method that uses a similarity measure for calculating the similarity between state-action pairs. Another work, which assumes a situation like that assumed in [4], is [15]. Authors of [15] instead of using a similarity measure used an Inverse reinforcement learning approach to approximate a reward function for shaping the potential function. Another technique that is presented for MARL is [9]. In [9] difference reward used to shape a potential function. The difference reward helps the agent to learn the impact of its action on the environment by not considering the impact of other agents' action on the environment.

4 Proposed potential based reward shaping method

Given these points and motivations, we go on to describe our approach to reinforce reward signal by using a knowledge gained from the learning process. So, we want a reward function that will change every time the agent make progress in the task. This reward function must encourage or punish the agent according to the best and worst episode until now. The reward function must also handle sparse reward signals. The function that represented in equation 12 has properties that we want. As we know from [14], changing reward function might change the optimal policy for the current task. Hence, we use reward shaping to manipulate the reward function in order to consider the agent's improvement during the learning process.

$$\phi(s, a, t) = \begin{cases} 0 & R(s, a) = 0 \\ 1 + \frac{R^{ep} - R_u^{ep}(t)}{R_u^{ep}(t) - R_l^{ep}(t)} & O.W \end{cases} \quad (12)$$

In equation 12: $R(s, a)$ is the immediate reward, which is a sparse reward signal; R^{ep} is the sum of rewards in the current episode, which we call it *episode reward*; $R_u^{ep}(t)$ is the maximum value of *episode reward* until now; and $R_l^{ep}(t)$ is the minimum value of *episode reward* until now. This function returns zero, while the reward signal of the environment has no information in it; the case to control the effect of sparse-reward. This function reinforces the reward signal by measuring a distance from a fixed point. We could use this approach with any kind of learning algorithm to boost the learning process. As presented in algorithm ??, after each episode of learning the agent will change parameters of potential function if needed.

4.1 Extending to multi-task agents

The presented potential function helps the learning methods to improve their performance by using knowledge extracted from previous episodes. Sometimes the agent needs to learn multiple tasks simultaneously. With this in mind, we extended the proposed method to be used in multi-task RL.

5 Results

In order to demonstrate the usefulness of the proposed approach, we perform experiments in two domains: Breakout and Pong games from arcade learning environment (ALE) [3]. According to the [3]:

在[5]中，RS被用作知识转移过程，并且从另一个任务中学到的策略被用作知识。

文献中还有许多其他作品提出了一种基于电位的多智能体强化学习（MARL）和单智能体强化学习（SARL）奖励塑造方法。[4]提出了RS的方法，同时有一个任务的演示。[4]提出了一种使用相似性度量来计算状态-动作对之间的相似性的方法。另一项工作是[15]，它假设了[4]中假设的情况。[15]的作者没有使用相似性度量，而是使用逆强化学习方法来近似奖励函数来塑造潜在函数。另一种针对 MARL 的技术是 [9]。在[9]中，差异奖励用于塑造潜在函数。差异奖励帮助智能体通过不考虑其他智能体的行为对环境的影响来了解其行为对环境的影响。

4 Proposed potential based reward shaping method

考虑到这些要点和动机，我们继续描述我们通过使用从学习过程中获得的知识来强化奖励信号的方法。因此，我们想要一个奖励函数，每当代理在任务中取得进展时，奖励函数就会发生变化。该奖励函数必须根据迄今为止最好和最差的情节来鼓励或惩罚智能体。奖励函数还必须处理稀疏奖励信号。方程 12 中表示的函数具有我们想要的属性。正如我们从[14]中知道的，改变奖励函数可能会改变当前任务的最优策略。因此，我们使用奖励塑造来操纵奖励函数，以考虑代理在学习过程中的改进。

$$\phi(s, a, t) = \begin{cases} 0 & R(s, a) = 0 \\ 1 + \frac{R^{ep} - R_u^{ep}(t)}{R_u^{ep}(t) - R_l^{ep}(t)} & O.W \end{cases} \quad (12)$$

式12中： $R(s, a)$ 是立即奖励，是稀疏奖励信号； R^{ep} 是当前剧集的奖励总和，我们称之为 *episode reward*； $R_u^{ep}(t)$ 是目前为止 *episode reward* 的最大值； $R_l^{ep}(t)$ 是迄今为止 *episode reward* 的最小值。该函数返回零，而环境的奖励信号中没有信息；控制稀疏奖励的影响的情况。该函数通过测量距固定点的距离来增强奖励信号。我们可以将这种方法与任何类型的学习算法结合使用来促进学习过程。如算法 ?? 中所示，在每次学习之后，代理将根据需要更改潜在函数的参数。

4.1 Extending to multi-task agents

所提出的潜在函数有助于学习方法通过使用从之前的片段中提取的知识来提高其性能。有时代理需要同时学习多个任务。考虑到这一点，我们将所提出的方法扩展到多任务强化学习中。

5 Results

为了证明所提出方法的有用性，我们在两个领域进行了实验：来自街机学习环境（ALE）的打砖块和乒乓球游戏[3]。根据[3]：

“ALE provides an interface to hundreds of Atari 2600 game environments, each one different, interesting, and designed to be a challenge for human players. ALE presents significant research challenges for reinforcement learning, model learning, model-based planning, imitation learning, transfer learning, and intrinsic motivation.”

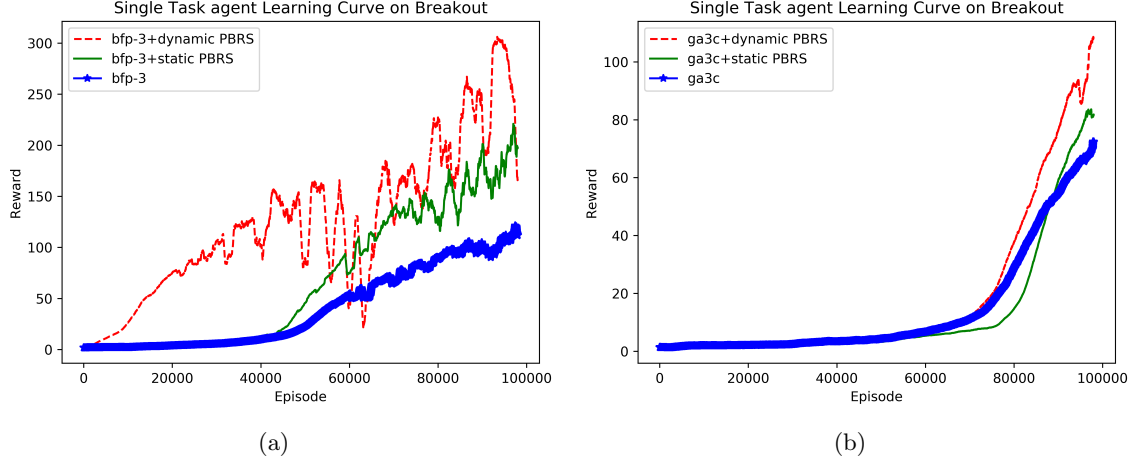


Figure 1: single task agent learning curve on Breakout

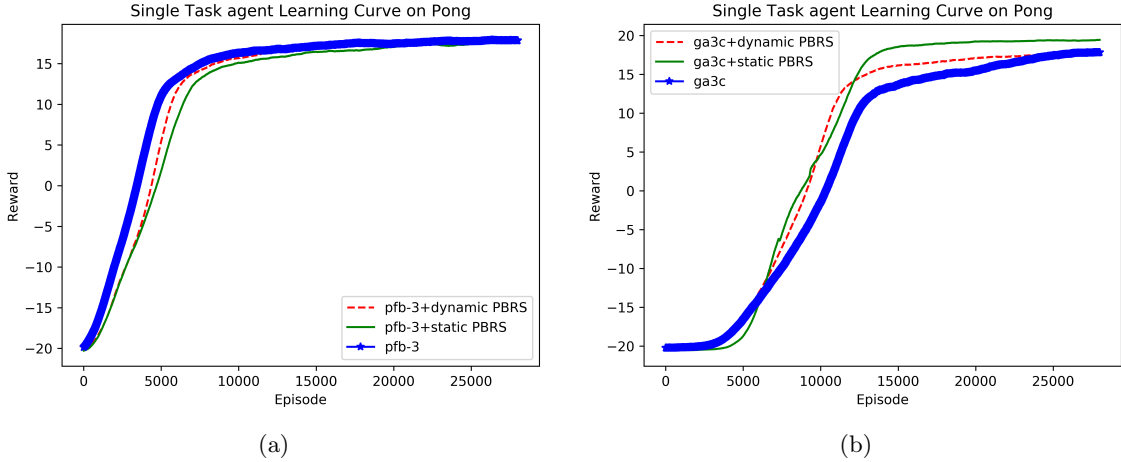


Figure 2: single task agent learning curve on Pong

We used two different baselines to compare our proposed method with them. [1] used as a first baseline and [6] as a second baseline. [1] is an implementation of [13], which uses a policy gradient method to approximate the optimal policy of a specific task. As a second baseline, we implement [6] that presented in the literature of transfer learning. We evaluate our work in two different phases: first of all, we evaluate our method during the learning process, and then we evaluate the

“ALE 提供了数百个 Atari 2600 游戏环境的界面，每个环境都不同、有趣，并且旨在为人类玩家带来挑战。ALE 为强化学习、模型学习、基于模型的规划、模仿学习、迁移学习和内在动机提出了重大的研究挑战。”

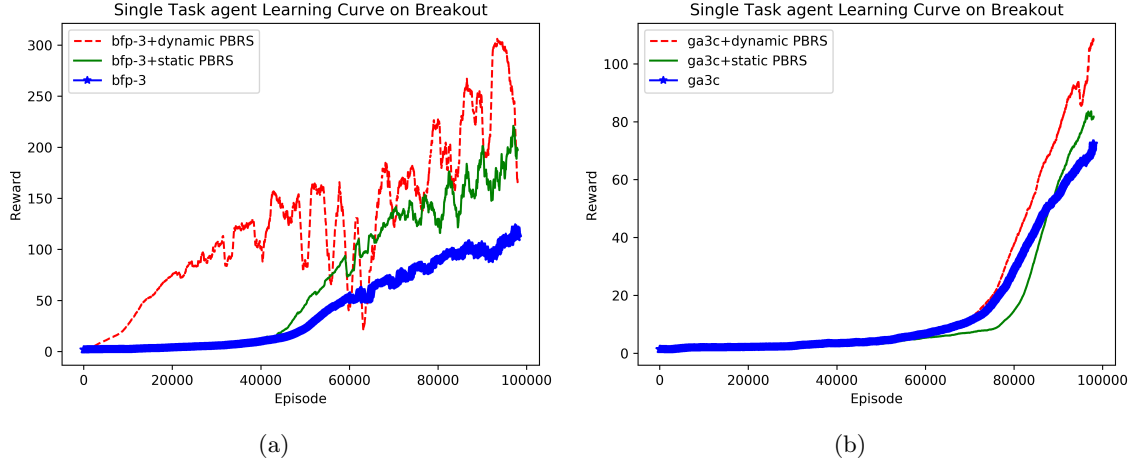


图 1: Breakout 上的单任务代理学习曲线

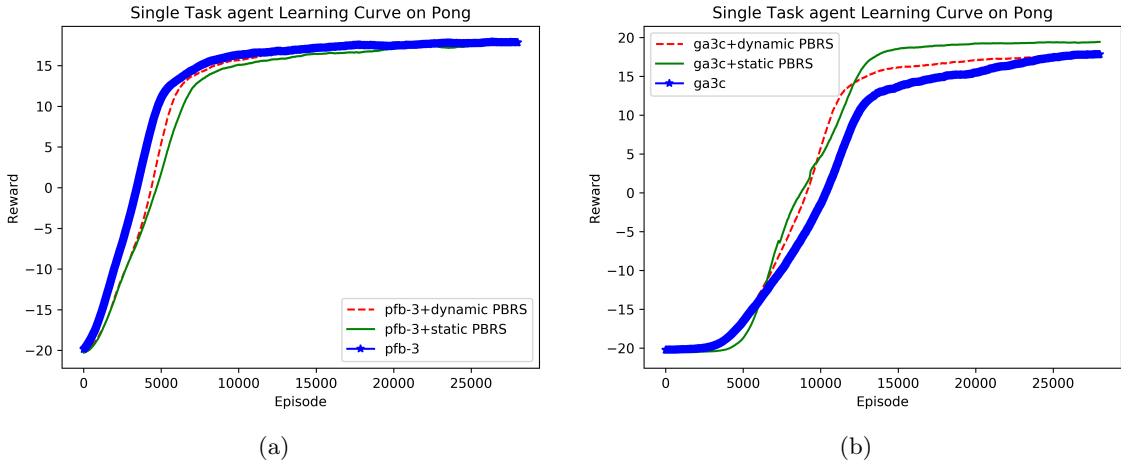


图 2: Pong 上的单任务代理学习曲线

我们使用两个不同的基线来将我们提出的方法与它们进行比较。[1]用作第一基线，[6]用作第二基线。[1]是[13]的实现，它使用策略梯度方法来近似特定任务的最优策略。作为第二个基线，我们实现了迁移学习文献中提出的[6]。我们分两个不同的阶段评估我们的工作：首先，我们在学习过程中评估我们的方法，然后我们评估

performance of the final policy. We tested our method with two different assumptions: the first assumption is that we know the maximum and minimum values of the episode reward in each task, and the second assumption is that we have no information about the episode reward of the task and this values will be obtained during the learning process.

5.1 Breakout

Breakout is an arcade game, which an agent handles a paddle to hit a ball trying to destroy more bricks while preventing the ball to cross the paddle. We trained an agent in this environment for 1000000 episodes. Figure 1 shows the learning curve of agents that trained on the Breakout game. As we see in figure 1 area under the curve of our method is greater than the baseline method, and the final performance is also better than the baseline.

5.2 Pong

Pong is also an arcade game, which an agent has the responsibility of moving a paddle to hit a ball. The agent will receive -1 reward if it loses the ball and will receive +1 if the opponent loses the ball. We trained an agent in this environment for 30000 episodes. Figure 2 shows the learning curve of agents that trained in the Pong game. As you see in figure 2 area under the curve of our method is greater than one of the baseline methods. However, in general, the proposed method in this environment has not been able to work very well and has a little improvement.

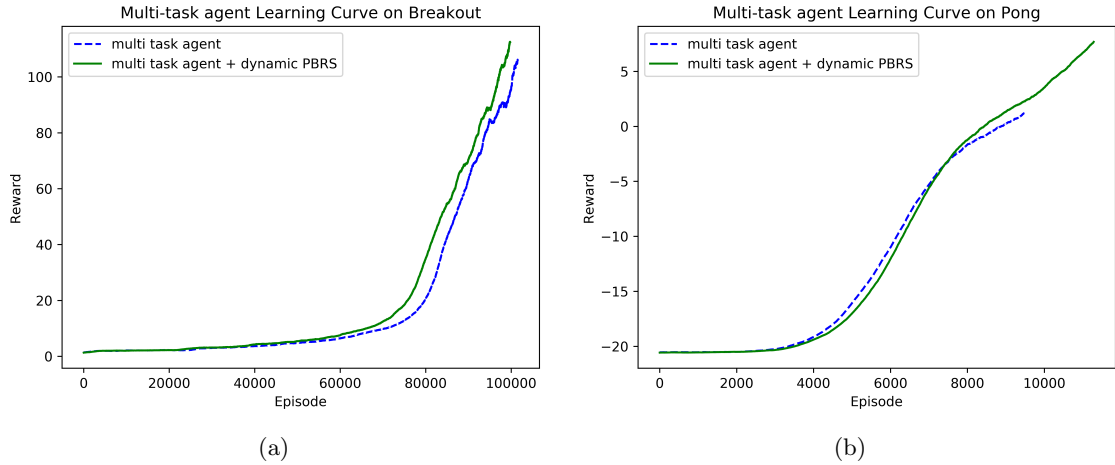


Figure 3: multi-task agent learning curve on Pong and Breakout

5.3 Multi-task Agent

We also implemented a multi-task agent, which learns how to play games of Pong and Breakout and then trains the agent for 115k episodes. Figure 3 and figure 4 demonstrate results of our experiments for a multi-task agent. Figure 3 demonstrates the learning curve of the multi-task agent for each of the games, while figure 4 demonstrates the performance of the learned-policy. Figure 4 is the result of running the agent with the learned-policy for 100 episode and then take the average rewards of

最终政策的执行情况。我们用两个不同的假设来测试我们的方法：第一个假设是我们知道每个任务中的情节奖励的最大值和最小值，第二个假设是我们没有关于任务的情节奖励的信息，并且该值将在学习过程中获得。

5.1 Breakout

Breakout 是一款街机游戏，特工手持球拍击打球，试图摧毁更多砖块，同时防止球越过球拍。我们在这个环境中训练了一个智能体 1000000 集。图 1 显示了在 Breakout 游戏中训练的智能体的学习曲线。如图 1 所示，我们的方法的曲线下面积大于基线方法，并且最终性能也优于基线。

5.2 Pong

Pong 也是一种街机游戏，代理人有责任移动球拍来击球。如果代理丢球，代理将获得 -1 奖励；如果对手丢球，代理将获得 +1 奖励。我们在这个环境中训练了一个智能体 30000 集。图 2 显示了在 Pong 游戏中训练的智能体的学习曲线。如图 2 所示，我们方法的曲线下面积大于基线方法之一。然而，总的来说，所提出的方法在这种环境下并不能很好地工作，并且有一点改进。

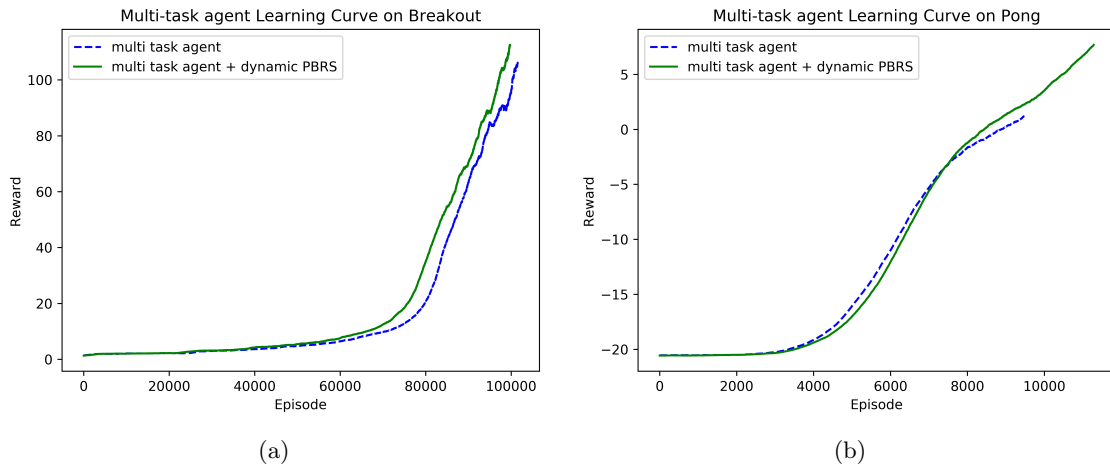


图 3: Pong 和 Breakout 上的多任务代理学习曲线

5.3 Multi-task Agent

我们还实现了一个多任务代理，它学习如何玩 Pong 和 Breakout 游戏，然后训练该代理 115k 集。图 3 和图 4 展示了我们的多任务代理实验结果。图 3 显示了每个游戏的多任务代理的学习曲线，而图 4 显示了学习策略的性能。图 4 是使用学习策略运行代理 100 集的结果，然后取平均奖励

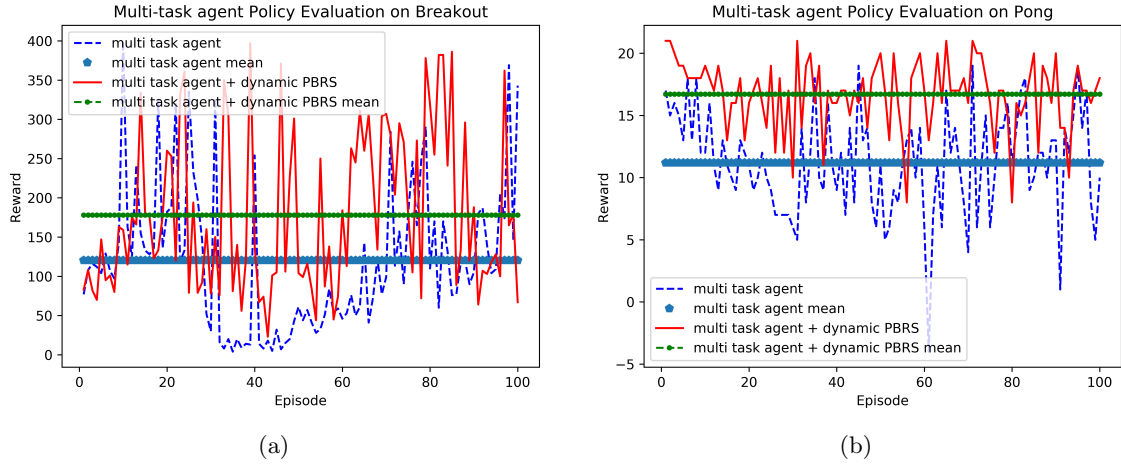


Figure 4: multi-task agent policy evaluation on Pong and Breakout

episodes. As we can see, the multi-task agent, which using reinforced reward signal, does perform better on average than a multi-task agent, which does not use reinforced reward signal.

6 Conclusion

First of all, in this paper, we studied the literature of transfer learning and potential-based reward shaping. According to our study in the literature of transfer learning and potential-based reward shaping, there was no method that tries to extract knowledge from the learning process. Hence, we introduced a novel way of extracting knowledge from the learning process, and we used reward shaping as a knowledge transferring method.

The proposed method guides the agent toward a goal by looking at the value of episode reward. If the agent goes toward the goal, the reward signal will be reinforced. This method can be used with any learning algorithm, and it supposed to be efficient wherever applied. We implemented our method using [1] and compare the results with two different baselines in two different environments, and then we extend our method to be used in multi-task agents. In most of the experiments, the results were promising. We also tested our method in a multi-task agent, which tries to learn games of Pong and Breakout simultaneously, and we saw that our method could lead to improvement.

References

- [1] Mohammad Babaeizadeh, Iuri Frosio, Stephen Tyree, Jason Clemons, and Jan Kautz. Reinforcement learning thorough asynchronous advantage actor-critic on a gpu. In *5th International Conference on Learning Representations*, 2017.
- [2] Peter L. Bartlett and Jonathan Baxter. Infinite-horizon policy-gradient estimation. *Journal of Artificial Intelligence Research*, 15(3):319–350, 2001.
- [3] Marc G. Bellemare, Yavar Naddaf, Joel Veness, and Michael Bowling. The arcade learning

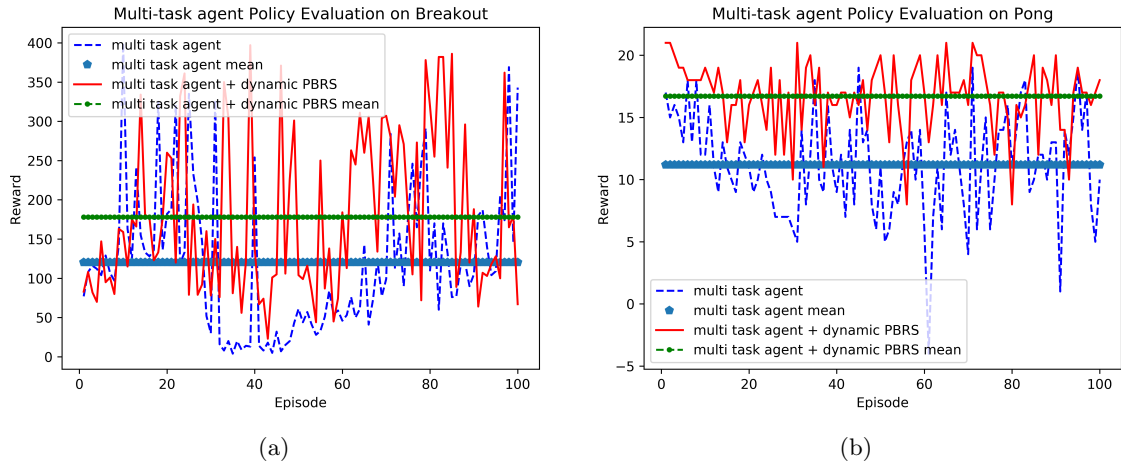


图 4: Pong 和 Breakout 上的多任务代理策略评估

剧集。正如我们所看到的，使用强化奖励信号的多任务代理的平均性能确实比不使用强化奖励信号的多任务代理更好。

6 Conclusion

首先，在本文中，我们研究了迁移学习和基于潜力的奖励塑造的文献。根据我们对迁移学习和基于潜力的奖励塑造文献的研究，没有任何方法试图从学习过程中提取知识。因此，我们引入了一种从学习过程中提取知识的新方法，并使用奖励塑造作为知识转移方法。

所提出的方法通过查看情节奖励的价值来引导智能体实现目标。如果智能体朝着目标前进，奖励信号就会得到加强。这种方法可以与任何学习算法一起使用，并且无论在哪里应用它都应该有效的。我们使用[1]实现了我们的方法，并将结果与两个不同环境中的两个不同基线进行比较，然后将我们的方法扩展到多任务代理中。在大多数实验中，结果都是有希望的。我们还在多任务代理中测试了我们的方法，该代理试图同时学习 Pong 和 Breakout 游戏，我们发现我们的方法可以带来改进。

References

- [1] 穆罕默德·巴巴伊扎德、尤里·弗罗西奥、斯蒂芬·泰里、杰森·克莱蒙斯和扬·考茨。通过 GPU 上的异步优势 Actor-Critic 进行强化学习。在 *5th International Conference on Learning Representations*, 2017 年。
- [2] 彼得·巴特利特和乔纳森·巴克斯特。无限范围政策梯度估计。 *Journal of Artificial Intelligence Research*, 15(3): 319–350, 2001。
- [3] Marc G. Bellemare、Yavar Naddaf、Joel Veness 和 Michael Bowling。街机学习

- environment: An evaluation platform for general agents. *J. Artif. Int. Res.*, 47(1):253–279, may 2013.
- [4] Tim Brys, Anna Harutyunyan, Halit Bener Suay, Sonia Chernova, Matthew E. Taylor, and Ann Nowé. Reinforcement learning from demonstration through shaping. In *Proceedings of the 24th International Conference on Artificial Intelligence*, IJCAI’15, pages 3352–3358. AAAI Press, 2015.
 - [5] Tim Brys, Anna Harutyunyan, Matthew E. Taylor, and Ann Nowé. Policy transfer using reward shaping. In *Proceedings of the 2015 International Conference on Autonomous Agents and Multiagent Systems*, AAMAS ’15, pages 181–188, Richland, SC, 2015. International Foundation for Autonomous Agents and Multiagent Systems.
 - [6] Gabriel De la Cruz, Yunshu Du, James Irwin, and Matthew Taylor. Initial progress in transfer for deep reinforcement learning algorithms. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2016.
 - [7] Sam Devlin and Daniel Kudenko. Dynamic potential-based reward shaping. In *Proceedings of the 11th International Conference on Autonomous Agents and Multiagent Systems - Volume 1*, AAMAS ’12, pages 433–440, Richland, SC, 2012. International Foundation for Autonomous Agents and Multiagent Systems.
 - [8] Sam Devlin and Daniel Kudenko. Plan-based reward shaping for multi-agent reinforcement learning. *Knowledge Eng. Review*, 31(1):44–58, 2016.
 - [9] Sam Devlin, Logan Michael Yliniemi, Daniel Kudenko, and Kagan Tumer. Potential-based difference rewards for multiagent reinforcement learning. In *AAMAS*, pages 165–172. IFAA-MAS/ACM, 2014.
 - [10] M. Grzes and D. Kudenko. Plan-based reward shaping for reinforcement learning. In *2008 4th International IEEE Conference Intelligent Systems*, volume 2, pages 10–22–10–29, Sept 2008.
 - [11] Marek Grzes and Daniel Kudenko. Multigrid reinforcement learning with reward shaping. In *Proceedings of the 18th International Conference on Artificial Neural Networks, Part I*, ICANN ’08, pages 357–366, Berlin, Heidelberg, 2008. Springer-Verlag.
 - [12] Anna Harutyunyan, Sam Devlin, Peter Vrancx, and Ann Nowe. Expressing arbitrary reward functions as potential-based advice. In *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence*, AAAI’15, pages 2652–2658. AAAI Press, 2015.
 - [13] Volodymyr Mnih, Adria Puigdomenech Badia, Mehdi Mirza, Alex Graves, Timothy Lillicrap, Tim Harley, David Silver, and Koray Kavukcuoglu. Asynchronous methods for deep reinforcement learning. In Maria Florina Balcan and Kilian Q. Weinberger, editors, *Proceedings of The 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1928–1937, New York, New York, USA, 20–22 Jun 2016. PMLR.
 - [14] Andrew Y. Ng, Daishi Harada, and Stuart J. Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the Sixteenth International Conference on Machine Learning*, ICML ’99, pages 278–287, San Francisco, CA, USA, 1999. Morgan Kaufmann Publishers Inc.

环境：针对总代理商的评估平台。 *J. Artif. Int. Res.*, 47(1): 253–279, 2013 年 5 月。

[4] Tim Brys、Anna Harutyunyan、Halit Bener Suay、Sonia Chernova、Matthew E. Taylor 和 Ann Nowé。从示范到塑造的强化学习。在 *Proceedings of the 24th International Conference on Artificial Intelligence, IJCAI' 15*, 第 3352-3358 页。美国人工智能出版社, 2015。

[5] Tim Brys、Anna Harutyunyan、Matthew E. Taylor 和 Ann Nowé。使用奖励塑造的政策转移。 *Proceedings of the 2015 International Conference on Autonomous Agents and Multiagent Systems, AAMAS '15*, 第 181-188 页, Richland, SC, 2015 年。国际自主代理和多代理系统基金会。

[6] 加布里埃尔·德拉克鲁兹、杜云舒、詹姆斯·欧文、马修·泰勒。深度强化学习算法迁移的初步进展。在 *International Joint Conference on Artificial Intelligence (IJCAI)*, 2016 年。

[7] 萨姆·德夫林和丹尼尔·库登科。基于动态潜力的奖励塑造。 *Proceedings of the 11th International Conference on Autonomous Agents and Multiagent Systems - Volume 1*, AAMAS '12, 第 433-440 页, Richland, SC, 2012 年。国际自主代理和多代理系统基金会。

[8] 萨姆·德夫林和丹尼尔·库登科。基于计划的多智能体强化学习奖励塑造。 *Knowledge Eng. Review*, 31 (1) : 44-58, 2016。

[9] Sam Devlin、Logan Michael Yliniemi、Daniel Kudenko 和 Kagan Tumer。多智能体强化学习的基于势的差异奖励。在 *AAMAS* 中, 第 165-172 页。IFAA-MAS/ACM, 2014 年。

[10] M. Grzes 和 D. Kudenko。基于计划的强化学习奖励塑造。 *2008 4th International IEEE Conference Intelligent Systems*, 第 2 卷, 第 10-22-10-29 页, 2008 年 9 月。

[11] Marek Grzes 和 Daniel Kudenko。具有奖励塑造的多重网格强化学习。 *Proceedings of the 18th International Conference on Artificial Neural Networks, Part I*, ICANN '08, 第 357-366 页, 柏林, 海德堡, 2008 年。Springer-Verlag。

[12] Anna Harutyunyan、Sam Devlin、Peter Vrancx 和 Ann Nowe。将任意奖励函数表达为基于潜力的建议。在 *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence, AAAI' 15*, 第 2652-2658 页。美国人工智能出版社, 2015。

[13] Volodymyr Mnih、Adria Puigdomenech Badia、Mehdi Mirza、Alex Graves、Timothy Lillicrap、Tim Harley、David Silver 和 Koray Kavukcuoglu。深度强化学习的异步方法。载于 Maria Florina Balcan 和 Kilian Q. Weinberger, 编辑, *Proceedings of The 33rd International Conference on Machine Learning, Proceedings of Machine Learning Research* 第 48 卷, 第 1928–1937 页, 纽约, 美国纽约, 2016 年 6 月 20–22 日。PMLR。

[14] 安德鲁·Y 吴、原田大师和斯图尔特·J 拉塞尔。奖励转型下的政策不变性：奖励塑造的理论与应用。 *Proceedings of the Sixteenth International Conference on Machine Learning, ICML '99*, 第 278-287 页, 美国加利福尼亚州旧金山, 1999 年。Morgan Kaufmann Publishers Inc.

- [15] Halit Bener Suay, Tim Brys, Matthew E. Taylor, and Sonia Chernova. Learning from demonstration for shaping through inverse reinforcement learning. In *Proceedings of the 2016 International Conference on Autonomous Agents & Multiagent Systems*, AAMAS '16, pages 429–437, Richland, SC, 2016. International Foundation for Autonomous Agents and Multiagent Systems.
- [16] Richard S. Sutton and Andrew G. Barto. *Introduction to Reinforcement Learning*. MIT Press, Cambridge, MA, USA, 1st edition, 1998.
- [17] Richard S. Sutton, David McAllester, Satinder Singh, and Yishay Mansour. Policy gradient methods for reinforcement learning with function approximation. In *Proceedings of the 12th International Conference on Neural Information Processing Systems*, NIPS'99, pages 1057–1063, Cambridge, MA, USA, 1999. MIT Press.
- [18] Matthew E. Taylor and Peter Stone. Transfer Learning for Reinforcement Learning Domains: A Survey. *JOURNAL OF MACHINE LEARNING RESEARCH*, 10:1633–1685, JUL 2009.
- [19] Christopher J.C.H. Watkins and Peter Dayan. Technical note: Q-learning. *Machine Learning*, 8(3):279–292, May 1992.
- [20] Eric Wiewiora, Garrison Cottrell, and Charles Elkan. Principled methods for advising reinforcement learning agents. In *Proceedings of the Twentieth International Conference on International Conference on Machine Learning*, ICML'03, pages 792–799. AAAI Press, 2003.
- [21] Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3):229–256, May 1992.

- [15] Halit Bener Suay、Tim Brys、Matthew E. Taylor 和 Sonia Chernova。通过逆强化学习从示范中学学习塑造。 *Proceedings of the 2016 International Conference on Autonomous Agents & Multiagent Systems*, AAMAS '16, 第 429-437 页, Richland, SC, 2016 年。国际自主代理和多代理系统基金会。
- [16] 理查德·S 萨顿和安德鲁·G 巴托。 *Introduction to Reinforcement Learning*。麻省理工学院出版社, 美国马萨诸塞州剑桥, 第 1 版, 1998 年。
- [17] 理查德·S 萨顿、大卫·麦卡莱斯特、萨蒂德·辛格和伊谢·曼苏尔。具有函数逼近的强化学习的策略梯度方法。 *Proceedings of the 12th International Conference on Neural Information Processing Systems*, NIPS' 99, 第 1057-1063 页, 美国马萨诸塞州剑桥, 1999 年。麻省理工学院出版社。[18] 马修·泰勒和彼得·斯通。强化学习领域的迁移学习: 一项调查。 *JOURNAL OF MACHINE LEARNING RESEARCH*, 10:1633-1685, 2009 年 7 月。
- [19] 克里斯托弗·J.C.H.沃特金斯和彼得·达扬。技术说明: Q-学习。 *Machine Learning*, 8(3): 279-292, 1992 年 5 月。
- [20] 埃里克·维奥拉 (Eric Wiewiora)、加里森·科特雷尔 (Garrison Cottrell) 和查尔斯·埃尔坎 (Charles Elkan)。为强化学习代理提供建议的原则方法。在 *Proceedings of the Twentieth International Conference on International Conference on Machine Learning*, ICML' 03, 第 792-799 页。AAAI出版社, 2003年。
- [21] 罗纳德·J 威廉姆斯。用于联结强化学习的简单统计梯度跟踪算法。 *Machine Learning*, 8(3): 229-256, 1992 年 5 月。